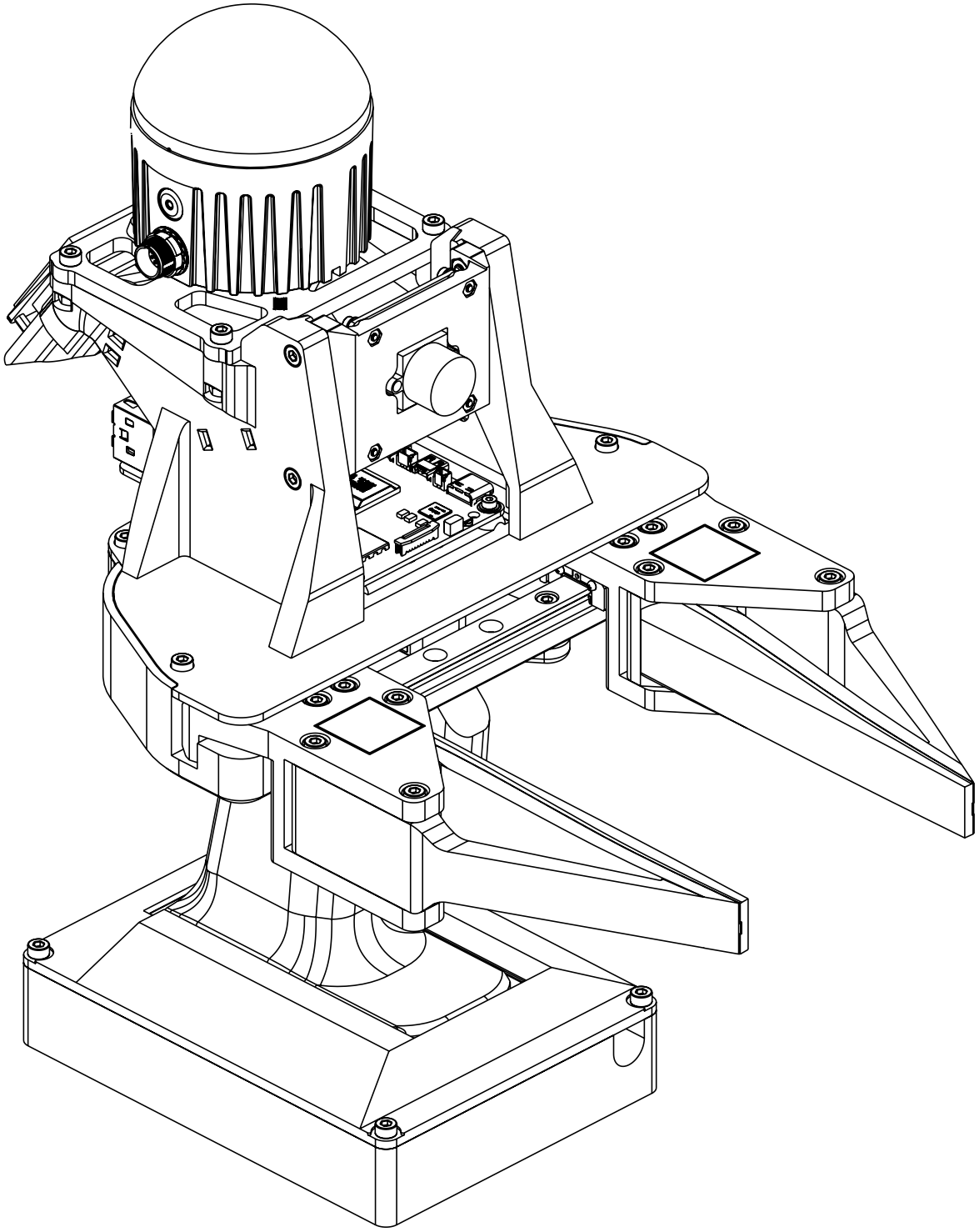
UMI-Live

Hardware Instructions



ASSEMBLY GUIDE

MECHANICAL / ELECTRONICS / WIRING

v6

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Body, trigger, and gears	6 - 11
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Notes

This guide assembles one UMI-Live gripper. Build two units for a bimanual setup.

Recommended Tools: M2/M3 drivers, scissors, flush cutters, multimeter, wire stripper.

3D Prints: remove and clean off all support materials.

Carriages: select two carriages that slide the smoothest on the rail.

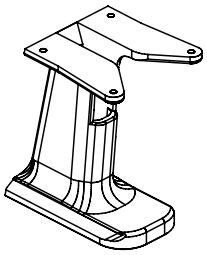
> **!** Keep carriages on a rail at all times to prevent the ball bearings from falling out.

Threadlocker: recommended for long-term or heavy use on metal-to-metal threads. Keep it off plastic, bearings, and sliding surfaces.

> **!** Keep the battery disconnected while assembling or changing wiring.

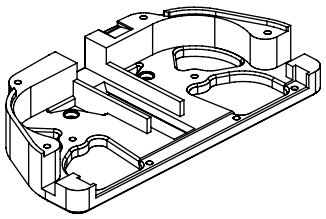
3D printed parts

grip



A **1x**

body_bottom



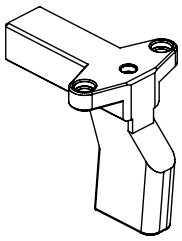
B **1x**

gear_axle



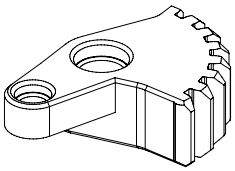
C **2x**

trigger



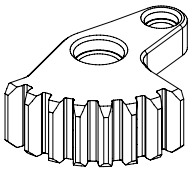
D **1x**

gear_left



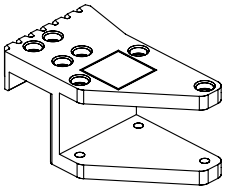
F **1x**

gear_right



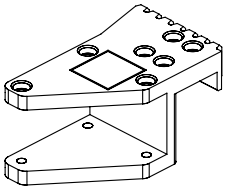
G **1x**

finger_holder_left



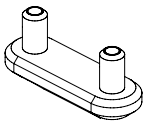
H-L **1x**

finger_holder_right



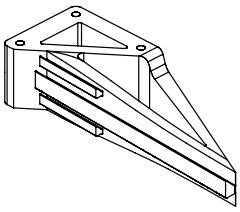
H-R **1x**

linkage



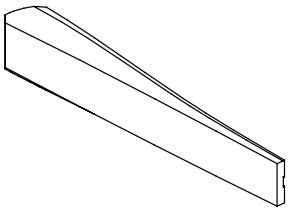
E **2x**

finger_backbone



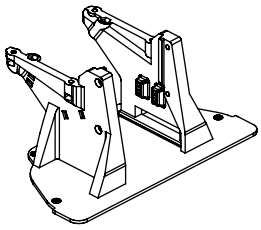
I **2x**

finger_pad



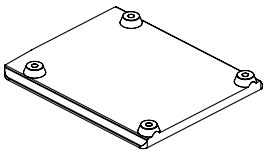
J **2x**

body_top



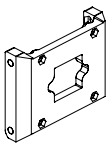
K **1x**

raspberry_slider



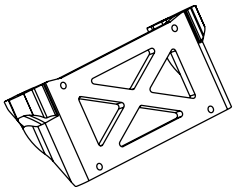
L **1x**

camera_mount



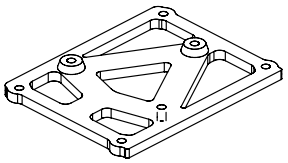
M **1x**

lcd_mount



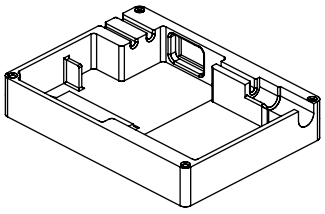
N **1x**

lidar_mount



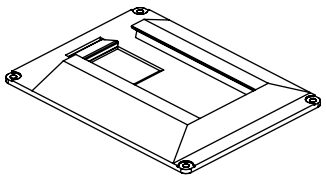
O **1x**

battery_pack_bottom



P **1x**

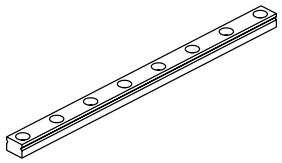
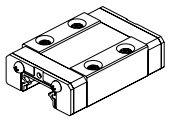
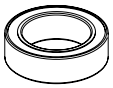
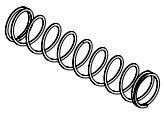
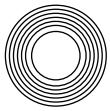
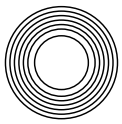
battery_pack_top



Q **1x**

Illustrations are not to scale.

Mechanical parts

MGN9 150mm Linear Rail  R **1x**	**MGN9C Carriage Blocks**  S **2x**	**uxcell MR85-2RS Ball Bearings**  T **8x**	**uxcell MR128-2RS Ball Bearings**  U **4x**
uxcell Compression Spring, 10x0.7x50mm  V **1x**	**ArUco tags (DICT_4X4_50, 16 mm)**  AM **2x**	**3M Gripping Material TB641, Black**  AN **Cut to fit**	**UHMW Polyethylene Transparent Film Slick Tape**  AP **Cut to fit**

Fasteners

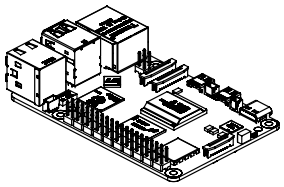
	H1 **M3 × 8 mm**	**14x**
	H2 **M3 × 12 mm**	**15x**
	H3 **M3 × 16 mm**	**2x**
	H4 **M3 × 20 mm**	**2x**
	H5 **M3 × 25 mm**	**10x**
	H6 **M3 × 30 mm**	**1x**
	H7 **M3 × 35 mm**	**6x**

	H8 **M3 nut**	**34x**
	H9 **M3 washer**	**2x**
	H10 **M2 × 6 mm**	**4x**
	H11 **M2 × 8 mm**	**8x**
	H12 **M2 nut**	**8x**
	H13 **M2 washer**	**12x**

Illustrations are not to scale.

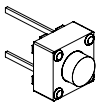
Electronics and cables

Raspberry Pi 5



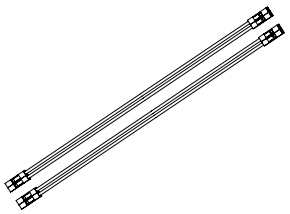
W **1x**

Tactile Button 2 Pin



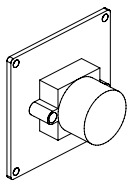
X **2x**

10cm Breadboard Jumper Wires



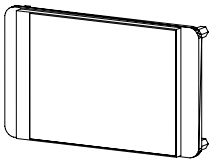
AL **4x**

SVPRO 2MP Global Shutter USB Camera



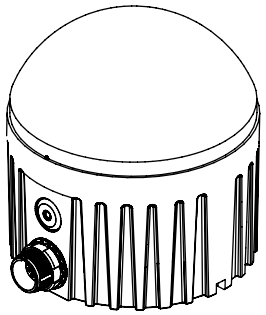
Y **1x**

Waveshare 2inch LCD Display SPI / 240 x 320



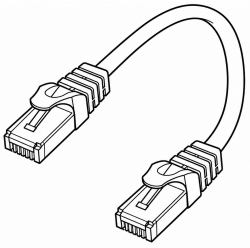
Z **1x**

RoboSense Airy LiDAR



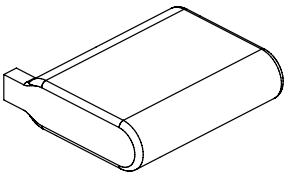
AA **1x**

Short Ethernet Cable 0.5FT



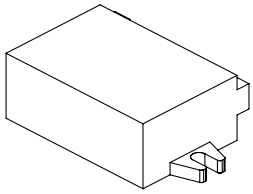
AK **1x**

12V 2600mAh Rechargeable Li-ion Battery Pack



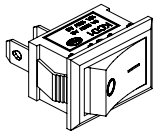
AB **1x**

DC-DC 12V/24V to 5V 5A 25W Buck Converter USB-C output



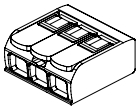
AC **1x**

DaierTek Mini ON and Off Rocker Switch



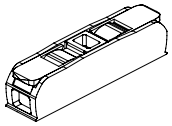
AD **1x**

WAGO 221-413 Lever Nuts



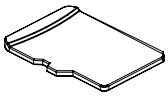
AF **2x**

WAGO 221-2401 Lever Nuts



AG **1x**

microSD card

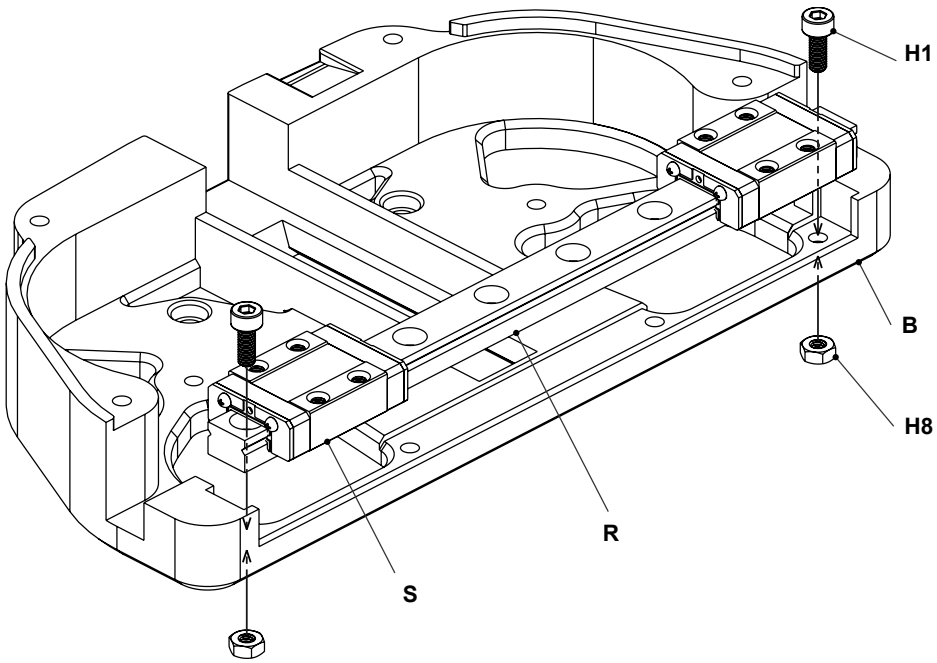


AH **1x**

Illustrations are not to scale.

STEP 1

Fit the rail and carriages



Select the two smoothest carriage blocks. Secure the rail to the body bottom using two M3 x 8 screws in the outer holes and two nuts underneath.

Keep both carriages on the rail to prevent ball bearings from falling out.

PARTS ADDED

B **1x**
Body bottom

R **1x**
Linear rail

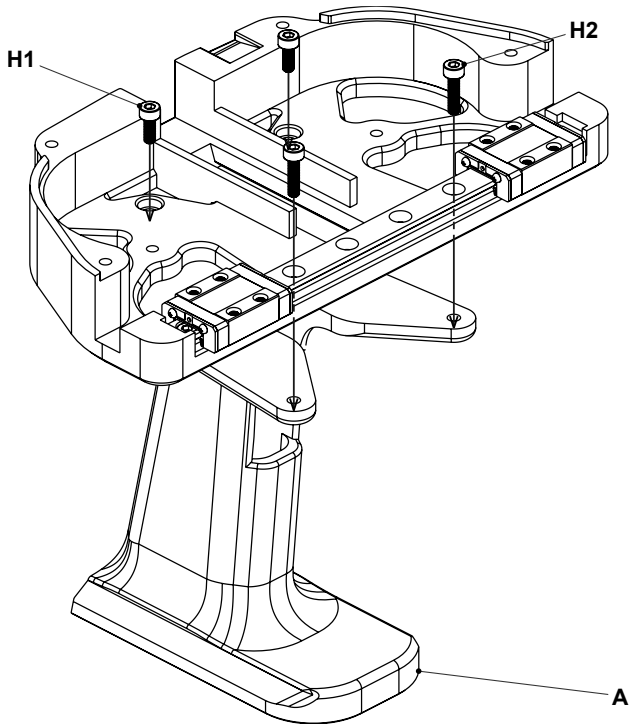
S **2x**
Carriage block

H1 **2x**
M3 × 8 mm

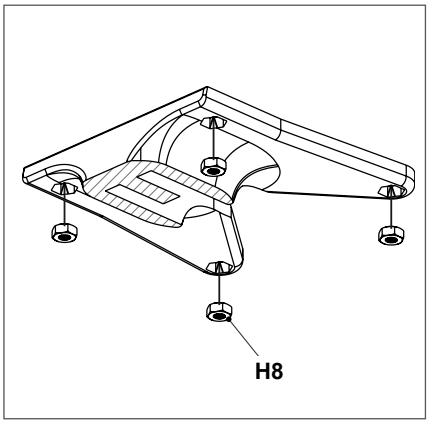
H8 **2x**
M3 nut

STEP 2

Attach the grip



UNDERSIDE VIEW



Align the grip beneath the body bottom. Fit two M3 x 12 screws through the inner rail holes, body bottom, and grip, plus two M3 x 8 screws through the other grip mounting holes. Secure all four screws with nuts underneath and tighten evenly.

PARTS ADDED

A **1x**
Grip

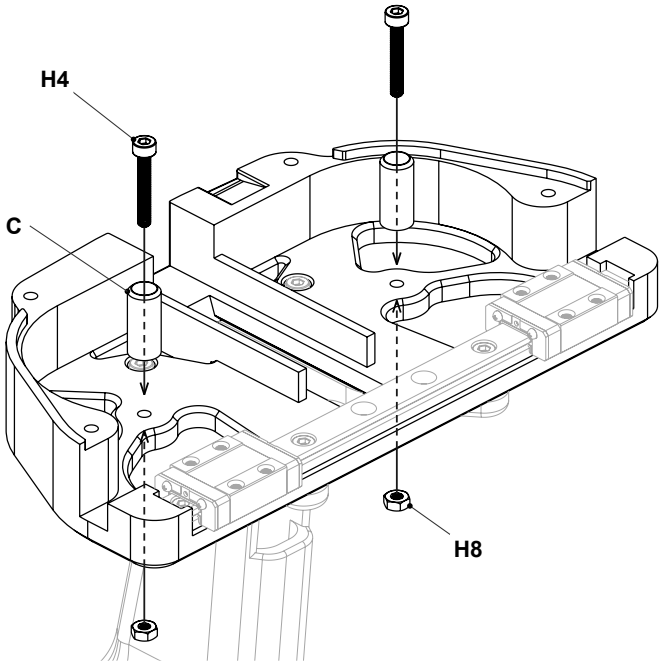
H1 **2x**
M3 × 8 mm

H2 **2x**
M3 × 12 mm

H8 **4x**
M3 nut

STEP 3

Install the two gear axles



Secure each gear axle with one M3 x 20 screw and nut. Ensure that both axles are seated vertically.

PARTS ADDED

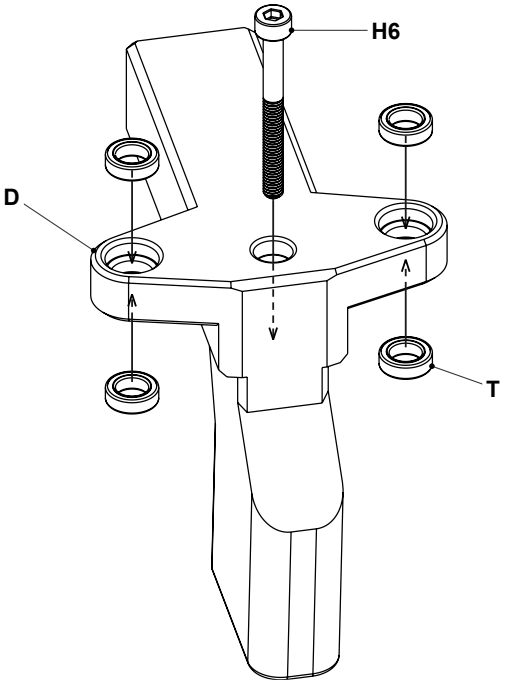
C **2x**
Gear axle

H4 **2x**
M3 × 20 mm

H8 **2x**
M3 nut

STEP 4

Prepare the trigger



Seat one small bearing on each face of both trigger pivots (four total). Screw the M3 x 30 screw into the central hole, threading the plastic.

Do not force the bearings into their recesses. If the fit is too tight, carefully file to adjust the trigger's bearing holes until the bearings seat without excessive force.

PARTS ADDED

D **1x**
Trigger

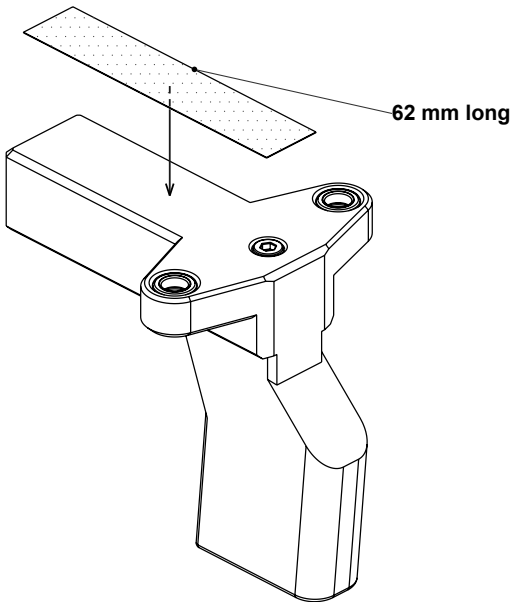
T **4x**
Small bearing

H6 **1x**
M3 × 30 mm

STEP 5

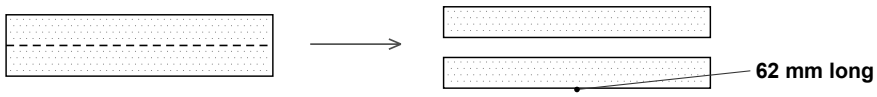
Apply low-friction tape

1. Top strip

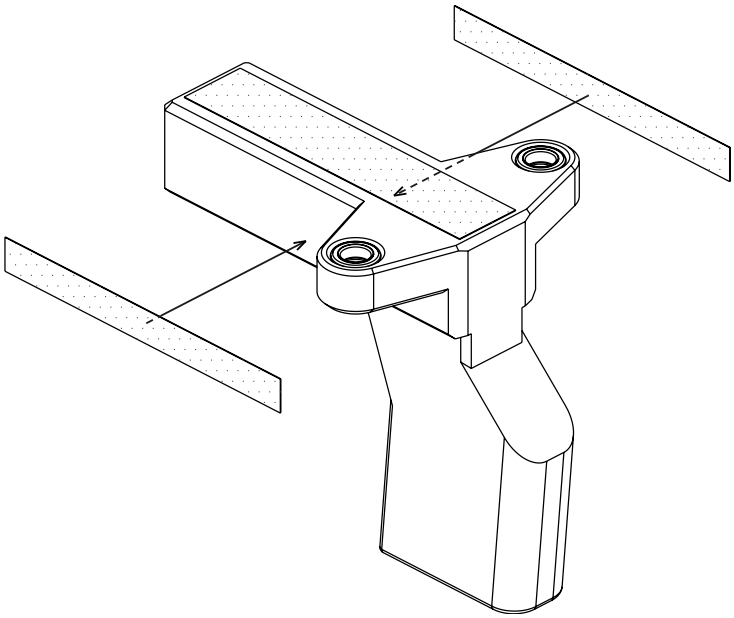


Cut a 62 mm long strip of low-friction tape. Apply it along the top of the trigger, covering the recessed screw.

2. Side strips



One half-width strip on each side



Cut another 62 mm long strip of low-friction tape, and cut it lengthwise. Apply one half-width strip to each side of the trigger.

Press firmly over all applied tape to secure it.

Do not stack or overlap tape layers. Apply tape only to flat surfaces; do not bend around corners.

PARTS ADDED

AP **Cut to fit**

Low-friction tape

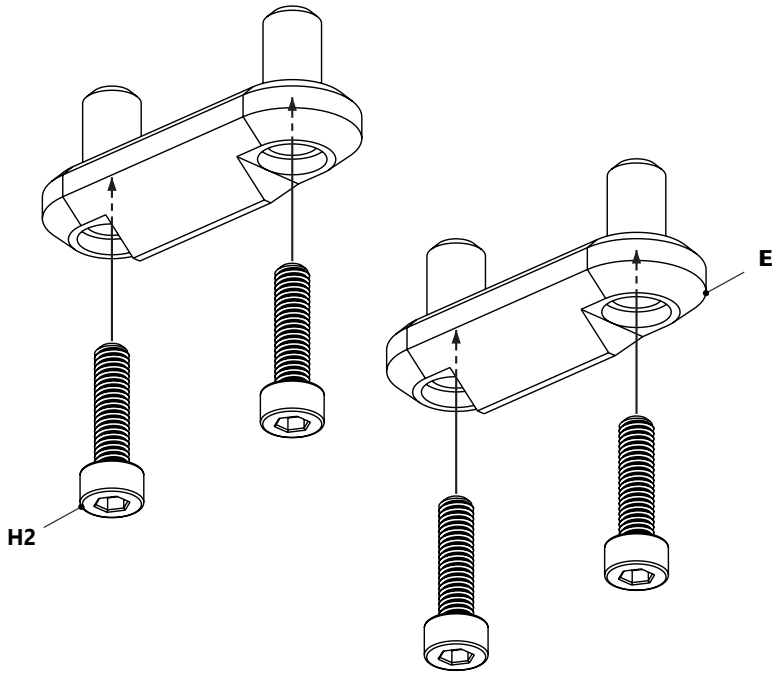
ALREADY ASSEMBLED

Prepared trigger

From Step 4

STEP 6

Prepare both linkages



Screw two M3 x 12 screws into each linkage from its underside, threading the plastic.

PARTS ADDED

E **2x**

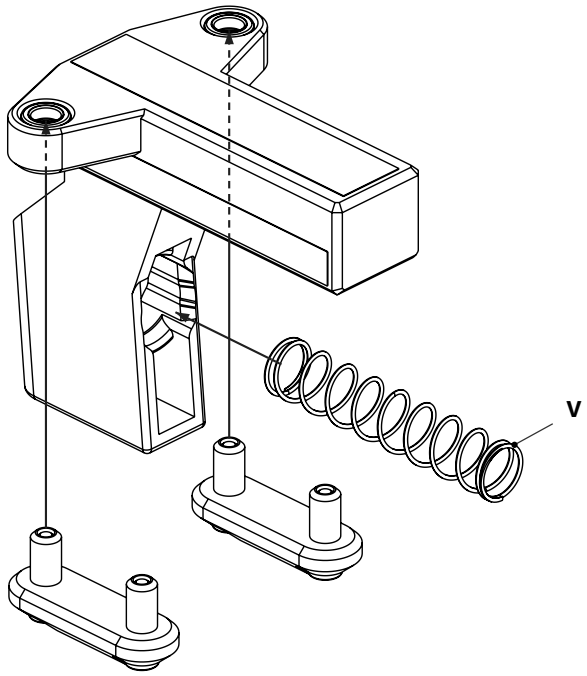
Linkage

H2 **4x**

M3 × 12 mm

STEP 7

Join the trigger and linkages



Bring both prepared linkages up from below into the trigger bearings. Slide the compression spring into the cylindrical slot in the trigger, along the axis shown.

PARTS ADDED

V **1x**

Compression spring

ALREADY ASSEMBLED

Prepared trigger

From Step 5

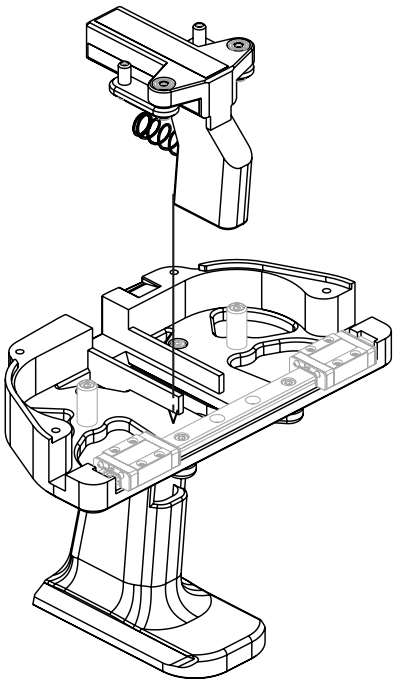
Prepared linkages

From Step 6

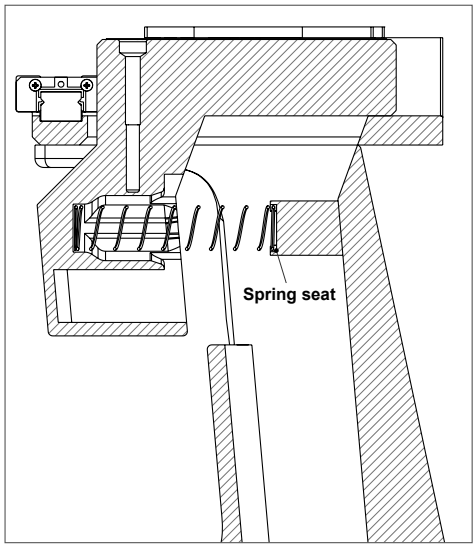
STEP 8

Seat the trigger in the body

LOWER INTO BODY



CROSS-SECTIONAL VIEW



Lower the trigger into its slot in the body assembly. Position the spring into its seat in the grip. Ensure both linkages remain fully seated in their bearings.

Tweezers may be needed to position the spring correctly in its seat.

ALREADY ASSEMBLED

Body assembly

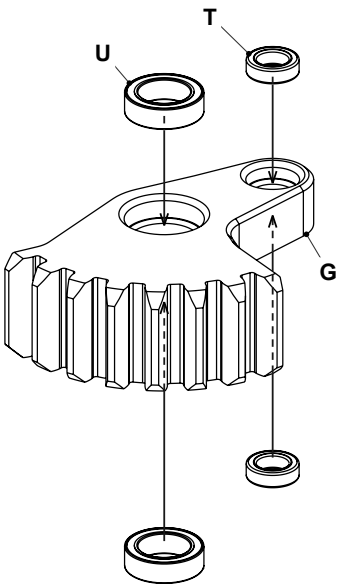
From Step 3

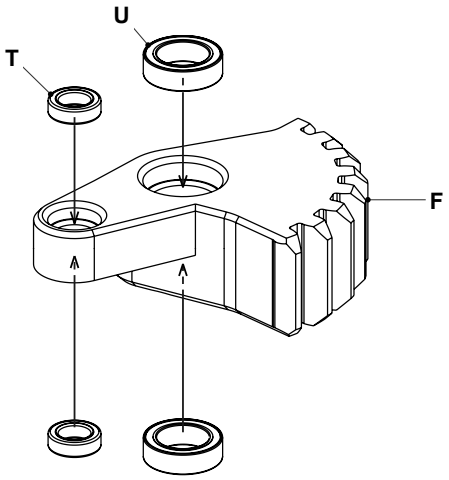
Trigger + linkages

From Step 7

STEP 9

Prepare the left and right gears





Fit two large MR128-2RS bearings into the main pivot and two small MR85-2RS bearings into the linkage pivot of each gear. Ensure that bearings are seated squarely in their recesses.

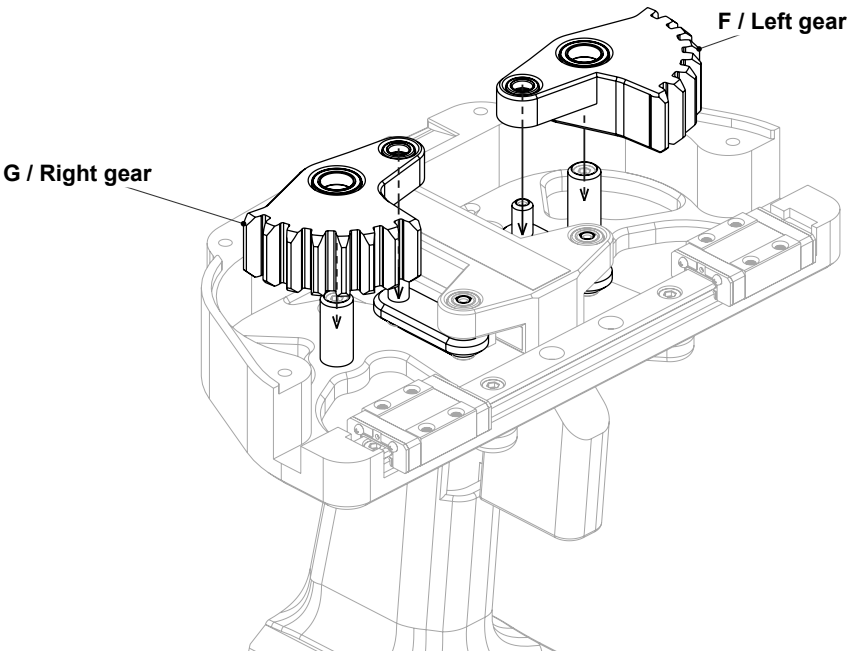
Do not force the bearings into their recesses. If the fit is too tight, carefully file to adjust the gear holes until the bearings seat without excessive force.

PARTS ADDED

F	**1x**
Left gear	
G	**1x**
Right gear	
U	**4x**
Large bearing	
T	**4x**
Small bearing	

STEP 10

Install the gears



Place the prepared gears over the two printed axles. Ensure that all bearings are fitted into their axles. Move the trigger gently to check that both sides travel together smoothly.

ALREADY ASSEMBLED

Body + trigger

From Step 8

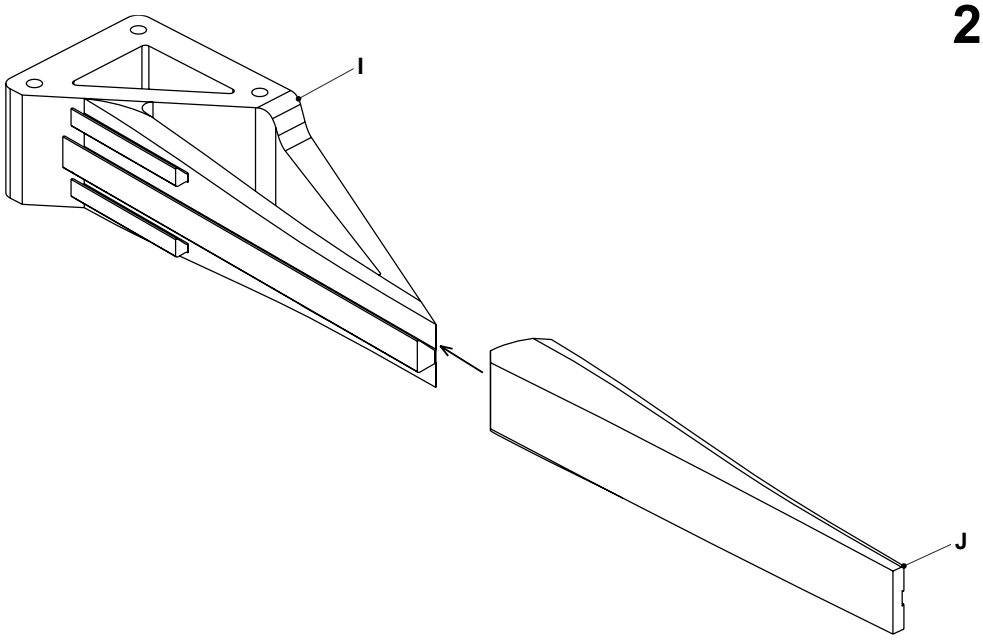
Prepared gears

From Step 9

STEP 11

Fit the TPU finger pads

2x



Slide the TPU pad onto its backbone from the finger tip. Keep it aligned with the backbone as you push/pull it along the full length. Build another set for a total of two fingers.

The pad slides on from the front; do not push it on sideways.

PARTS ADDED

I **2x**

Finger backbone

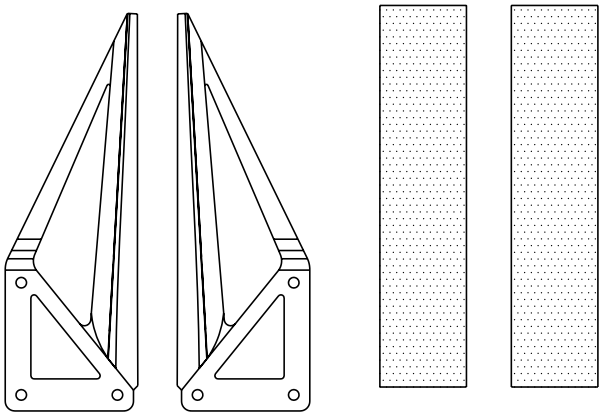
J **2x**

Finger pad

STEP 12

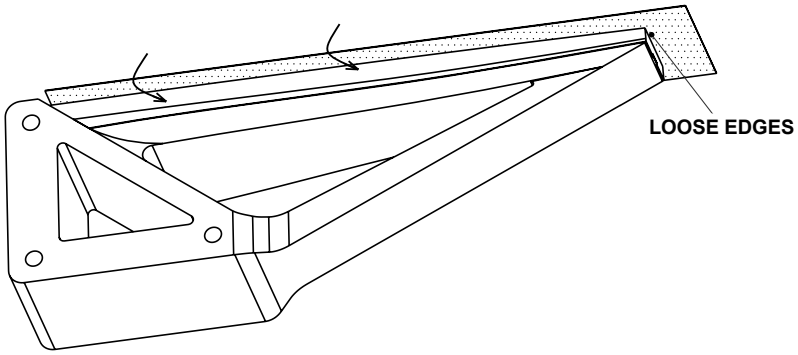
Apply the grip tape

1. Ready for tape



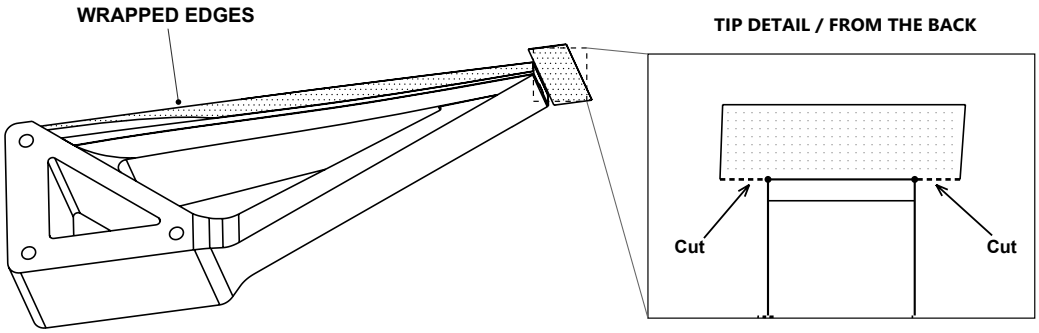
Start with both padded fingers from Step 11. Cut one tape strip for each, leaving enough to wrap the long edges and tip.

2. Apply the tape / leave the tip long



Center the strip along the TPU contact face. Leave tape overhanging both long edges and the finger tip.

3. Free the end tab / wrap the long edges



Cut inward from each tape edge to the tip corner (detail). Leave the center uncut, then smooth the long edges around the pad.

Continue with the enlarged tip views on the next page. Cut only the tape; keep the TPU pad and printed backbone intact.

PARTS ADDED

AN **Cut to fit**

Grip tape

ALREADY ASSEMBLED

Finger pair

From Step 11

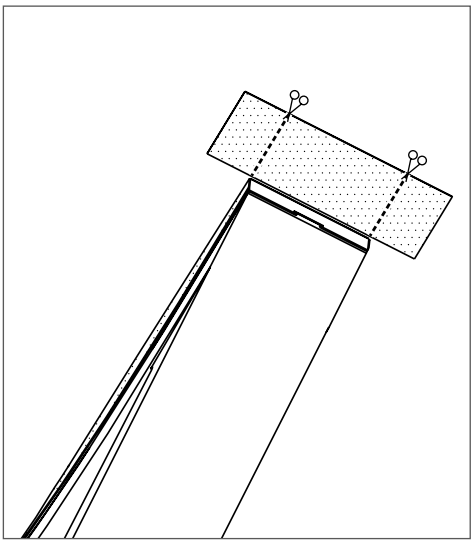
STEP 12 CONT.

Finish the grip tape

CONTINUE WITH

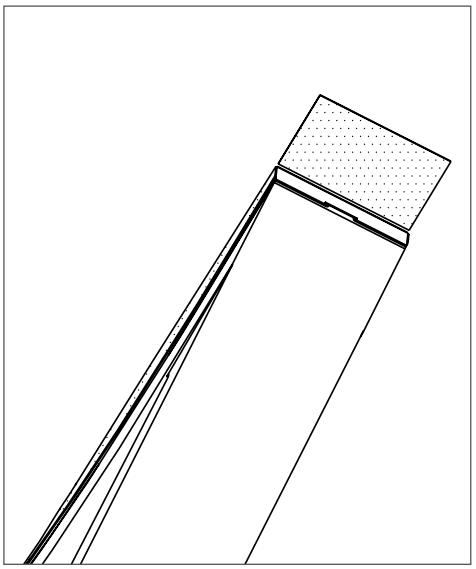
Same fingers and grip tape from the previous page.

4. Wide tab / trim the sides



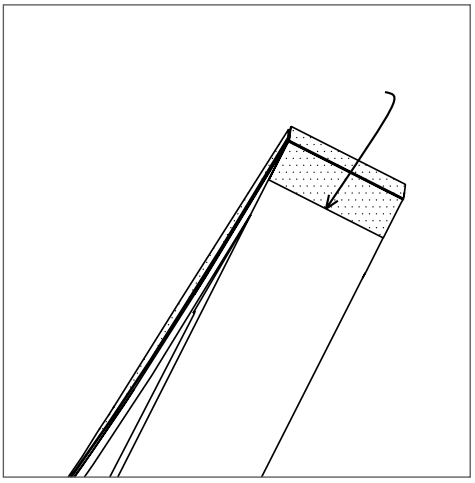
View the flat back of the tip. Trim the tab sides along the dashed lines, aligned with the two edges of the white backbone.

5. Tab trimmed to width



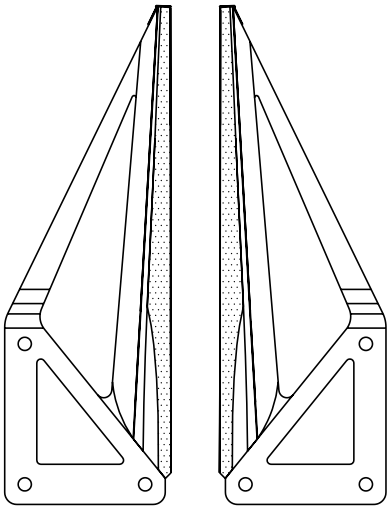
The tab now matches the flat back of the finger. Keep this center flap long enough to fold over the tip.

6. Fold flat over the tip



Fold the flap over the tip and press it flat against the back of the finger.

7. Finished pair



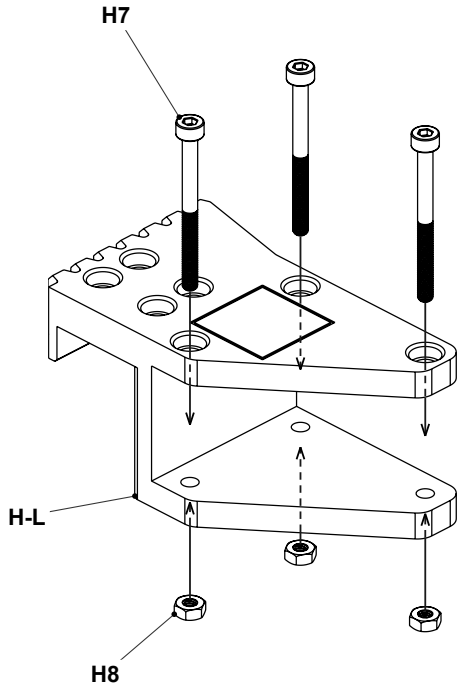
Repeat for the other finger.

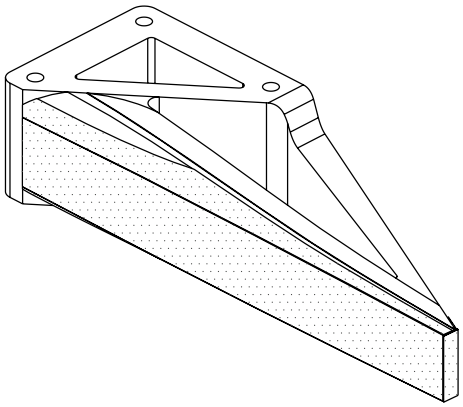
Press all tape edges and the folded tip firmly into place.
Trim any loose material clear of the mechanism.

STEP 13

Fit a finger to its holder

2x





Slide one completed finger into the left holder (H-L), shown, from the finger-tip side. Align the three holes and fit three M3 × 35 mm fasteners and three nuts. Attach one ArUco tag in the square recess. Repeat with the right holder (H-R) and the other finger.

Build one H-L and one H-R assembly. See tag sizing and IDs on page 15.

PARTS ADDED

H-L **1x**
Left finger holder

H-R **1x**
Right finger holder

H7 **6x**
M3 × 35 mm

H8 **6x**
M3 nut

AM **2x**
ArUco tags

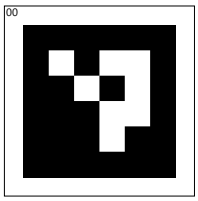
ALREADY ASSEMBLED

Finger pair
From Step 12

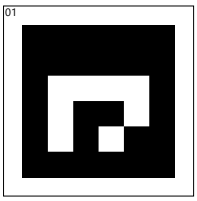
Fit the tags

Each unit uses two tags, one within each recessed border on the finger holders. Print the supplied ArUco tag sheet at **Actual size / 100%**, with Fit and Shrink disabled. Verify the 50 mm scale check before cutting.

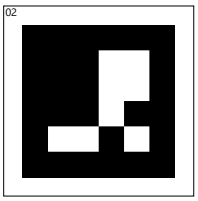
SETUP	TAG IDS
Each UMI-Live	Two distinct IDs
Left / right assignment	Configurable in software



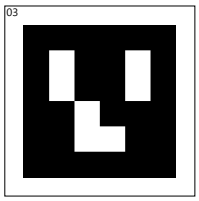
ID 0



ID 1



ID 2



ID 3

TAG PATTERNS SHOWN FOR IDENTIFICATION; PRINT FROM THE SUPPLIED TAG SHEET.

Size and placement

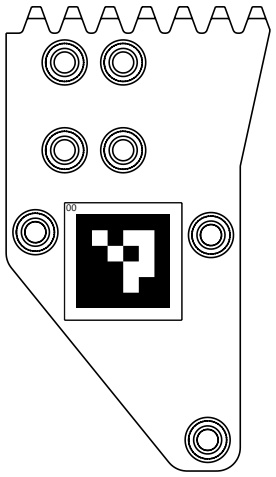
Dictionary: DICT_4X4_50.

Black marker: 16 x 16 mm, including its black border.

Cut size: 20 x 20 mm; retain the 2 mm white margin.

Attachment: tape or glue.

Keep the printed face flat, clean and visible.



TAG WITHIN RECESSED BORDER

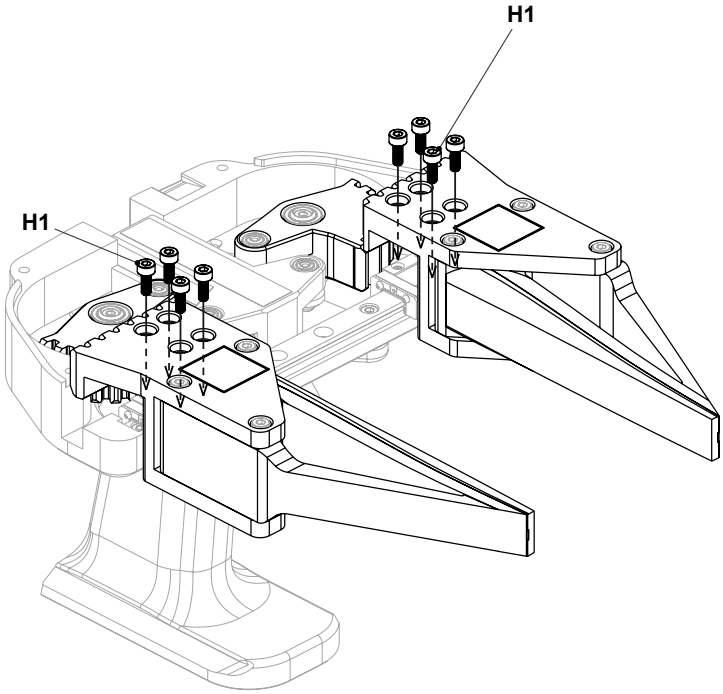
Tag orientation does not matter. The tag IDs for each UMI-Live gripper can be configured in software.

Fit each tag within the recessed border without covering the white margin. Keep the printed face flat, clean and visible.

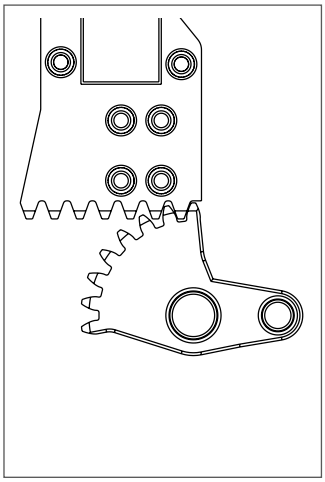
Print from ArUco_tags_0-3_all_16mm_US_Letter.pdf (IDs 0-3).
Do not cut tags from the enlarged diagrams in this manual.

STEP 14

Attach both finger assemblies



TOP VIEW / LEFT SIDE



F meshes with H-L

Mesh the left holder (H-L) with the left gear (F), and the right holder (H-R) with the right gear (G). Align each holder over its carriage and secure it with four M3 × 8 mm fasteners. Keep the padded faces facing each other and check that both fingers move symmetrically.

PARTS ADDED

H1 **8x**

M3 × 8 mm

ALREADY ASSEMBLED

Body + gears

From Step 10

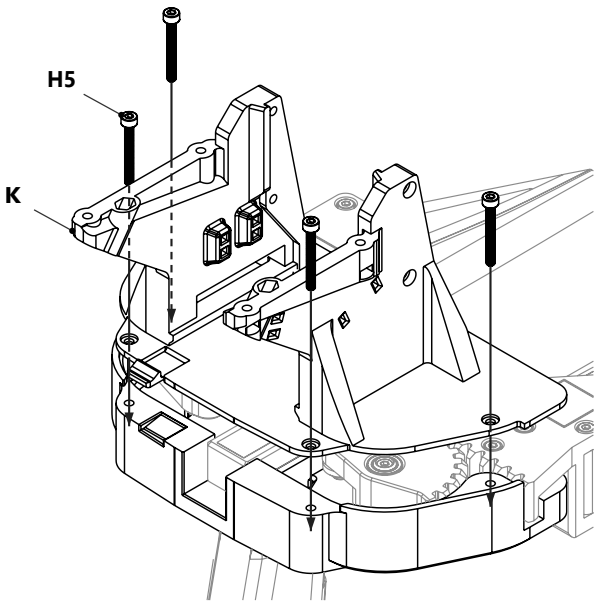
Finger assemblies

From Step 13

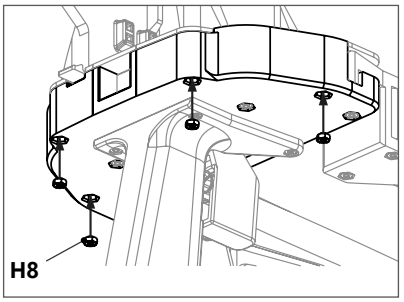
STEP 15

Close the body

BODY CLOSURE / CROPPED VIEW



NUT POCKETS / UNDERSIDE



Lower the body top over the completed mechanism. Fit four M3 x 25 screws and nuts. Tighten gradually, checking trigger return and free carriage motion as you go.

PARTS ADDED

K **1x**

Body top

H5 **4x**

M3 × 25 mm

H8 **4x**

M3 nut

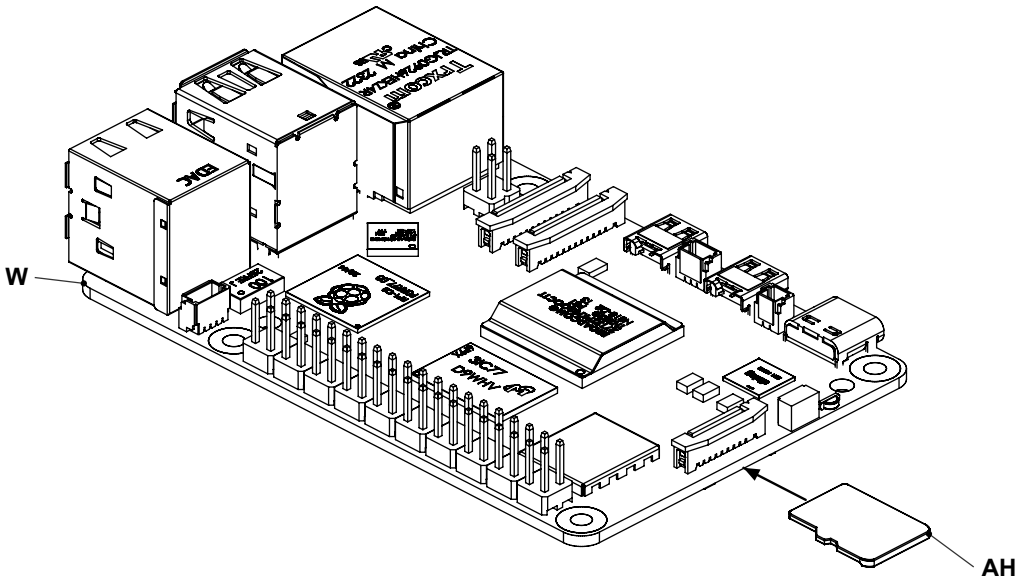
ALREADY ASSEMBLED

Body + fingers

From Step 14

STEP 16

Prepare the Pi



Insert the microSD card into the underside slot, contact end first, with the gold contacts facing the board and the label facing away.

Slide it in gently; do not force it. Prepare the microSD card using the UMI-Live software setup guide before first use. Software setup can be completed after assembly.

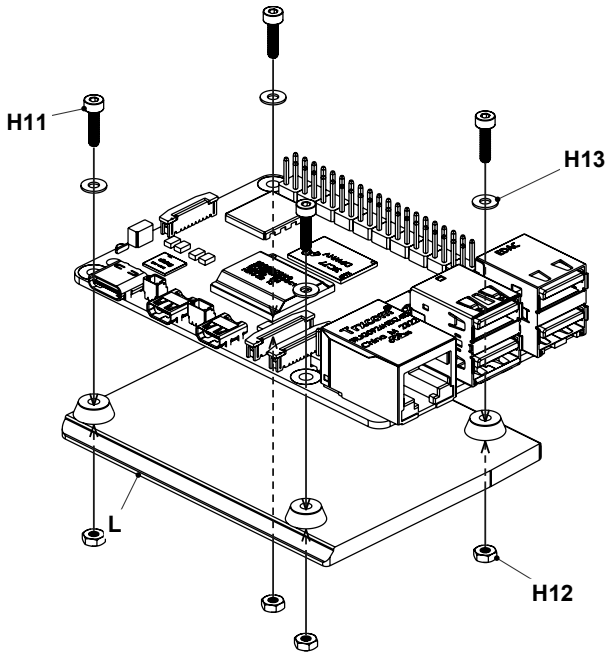
PARTS ADDED

W **1x**
Raspberry Pi 5

AH **1x**
microSD card

STEP 17

Mount the Pi on its slider



Place the Pi on its slider and align the four mounting holes. Fit four M2 x 8 screws, washers and nuts. Tighten just enough to hold the PCB without bending it.

Place each washer directly under its screw head, against the PCB.

PARTS ADDED

L **1x**
Pi slider

H11 **4x**
M2 × 8 mm

H12 **4x**
M2 nut

H13 **4x**
M2 washer

ALREADY ASSEMBLED

Prepared Pi
From Step 16

LCD + button pin map

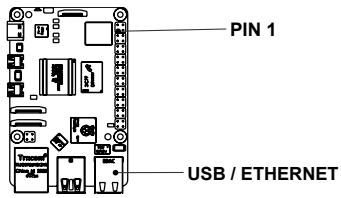
GPIO numbers use **BCM numbering**; pins use physical numbering. With the Pi **component side up**, orient the USB/Ethernet ports at the bottom and the header on the right. Pin 1 is the header's upper-left pin, farthest from those ports. Keep power disconnected.

LCD	PI SIGNAL	PIN
VCC	3.3V	1
GND	Ground	6
DIN	GPIO10 / MOSI	19
CLK	GPIO11 / SCLK	23
CS	GPIO8 / CE0	24
DC	GPIO25	22
RST	GPIO27	13
BL	GPIO18	12

Button connections

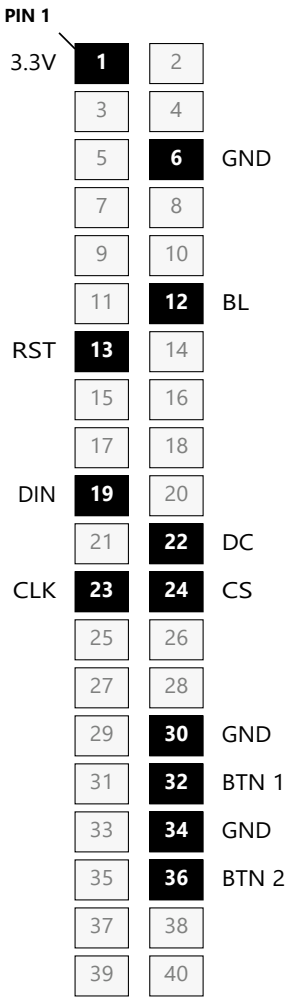
BUTTON	SIGNAL	GROUND
BTN 1 Near fingers	GPIO12 / pin 32	Pin 30
BTN 2 Near LCD	GPIO16 / pin 36	Pin 34

COMPONENT SIDE UP



40-PIN HEADER

COMPONENT SIDE / PHYSICAL PINS



USB / ETHERNET END OF BOARD

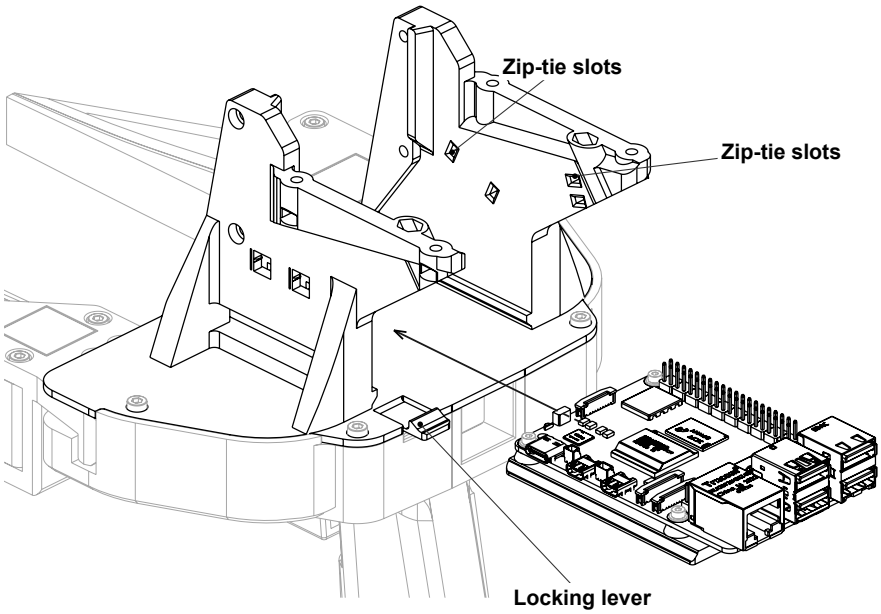
> **!** GPIO12 is physical pin 32. Physical pin 12 is LCD backlight GPIO18.

Buttons: use female-to-female leads. One terminal goes to the specified GPIO; the other goes to its paired ground. All four leads occupy the even-numbered row at pins 30, 32, 34 and 36.

LCD: follow the signal labels, rather than cable colors. Use Pi 3.3V for VCC in this build. The camera uses USB 3.0; Airy uses Ethernet and separate power.

STEP 18

Slide the Pi into the body



Slide the Pi and slider forward into the body top guides until the lever clicks and locks. Thread optional zip ties through the marked slots for cable management; leave them loose until wiring is complete.

PARTS ADDED

2 optional

Zip ties

ALREADY ASSEMBLED

Body assembly

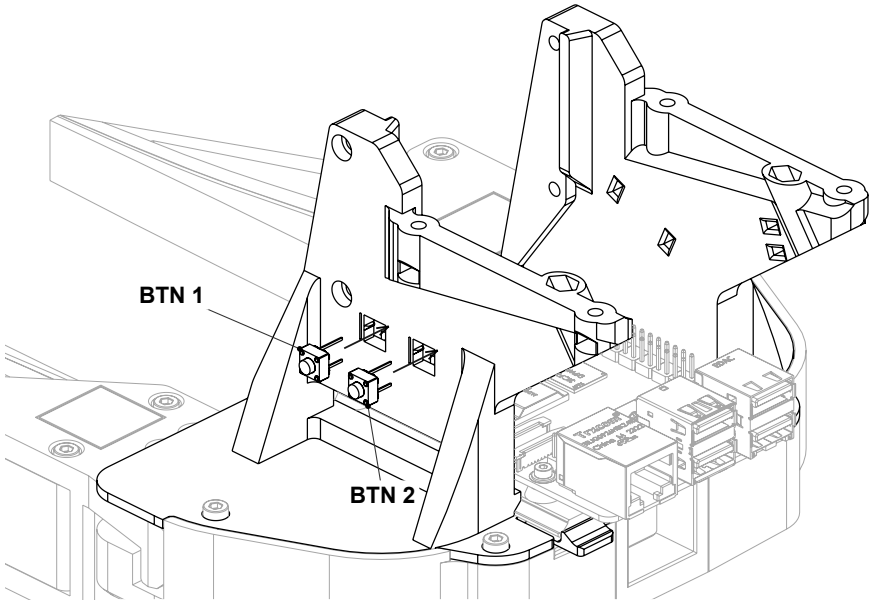
From Step 15

Pi + slider

From Step 17

STEP 19

Install the two buttons



Seat the buttons in the side openings. Connect the button nearest the fingers to GPIO12 and GND (pins 32 and 30). Connect the button nearest the LCD to GPIO16 and GND (pins 36 and 34).

Use four female-to-female leads; connect both ends using page 18. Push each socket fully onto its terminal and gently tug-check the connection.

PARTS ADDED

X **2x**

Tactile button

AL **4x**

Button leads

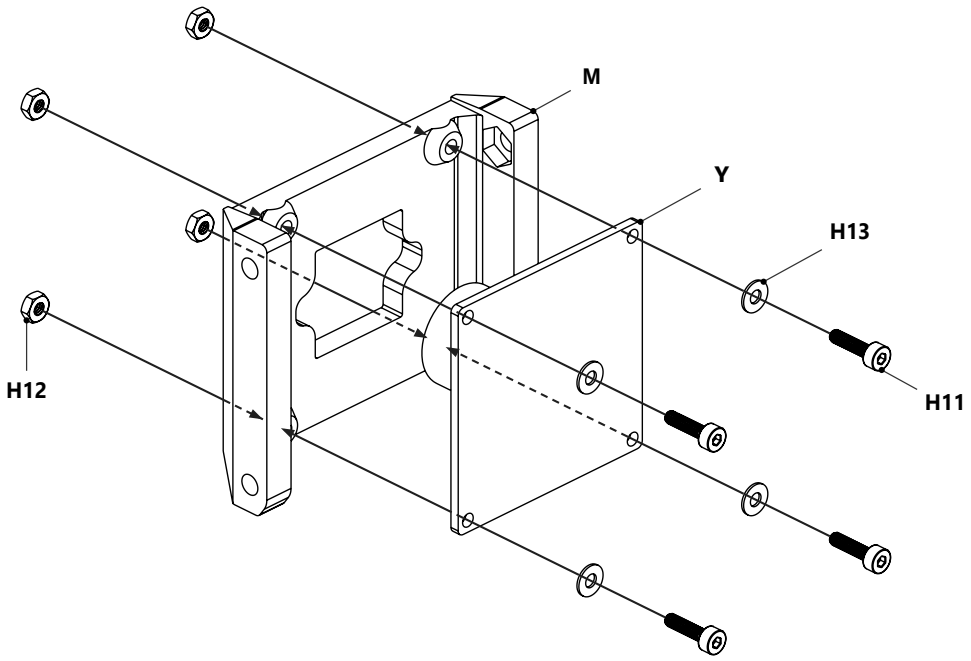
ALREADY ASSEMBLED

Body + Pi

From Step 18

STEP 20

Mount the camera board



Bring the camera board forward from behind the mount, with the lens facing through the opening. Fit four M2 x 8 screws, washers and nuts. Connect the 4-pin JST PH end of the USB camera cable to the camera board.

Washers go directly under the screw heads. Keep the lens and PCB clear.

PARTS ADDED

M **1x**
Camera mount

Y **1x**
SVPRO USB camera

From camera **1x**
USB Camera Cable

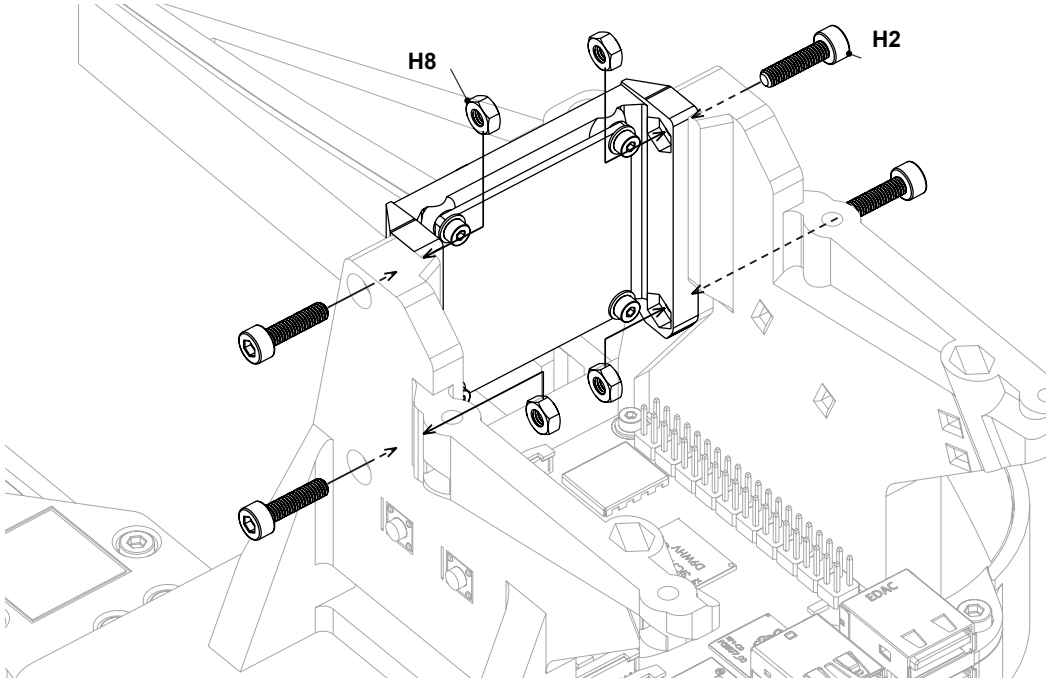
H11 **4x**
M2 × 8 mm

H12 **4x**
M2 nut

H13 **4x**
M2 washer

STEP 21

Attach and connect the camera



Secure the camera mount to the front of the body top with four M3 x 12 screws and nuts. Plug the USB camera cable into a blue USB 3.0 Type-A port on the Raspberry Pi.

PARTS ADDED

H2 **4x**
M3 × 12 mm

H8 **4x**
M3 nut

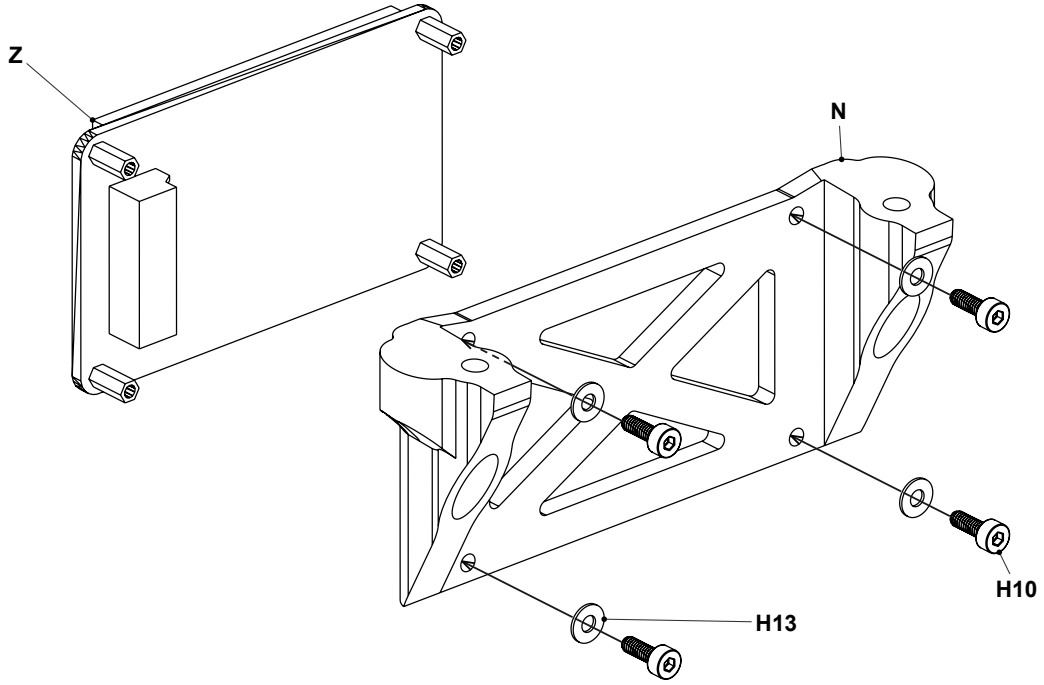
ALREADY ASSEMBLED

Body + buttons
From Step 19

Camera assembly
From Step 20

STEP 22

Fit the LCD to its mount



Place the LCD in its printed mount with the display facing outward. Fit four M2 x 6 screws and washers at the mounting holes. Keep the display cable and connector clear of the frame.

The four washers sit under the M2 screw heads.

PARTS ADDED

N **1x**
LCD mount

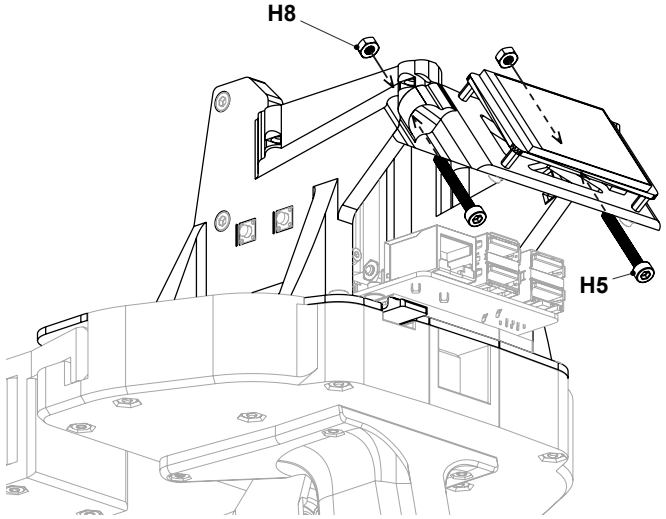
Z **1x**
Waveshare LCD

H10 **4x**
M2 × 6 mm

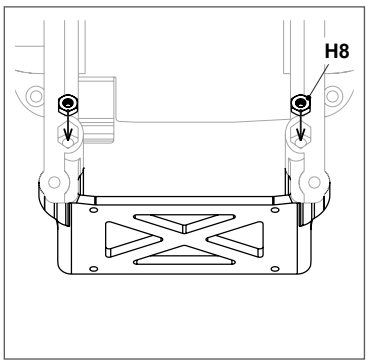
H13 **4x**
M2 washer

STEP 23

Attach and connect the LCD



TOP VIEW / NUT SLOTS



LCD omitted for clarity

Mount the LCD assembly at the rear of the body using two M3 x 25 screws and nuts. Connect the eight-wire LCD cable to the LCD and Pi header using the pin map on page 18. Recheck every signal before applying power.

Ensure that the LCD cable is fully pushed into the LCD display.

PARTS ADDED

H5 **2x**
M3 × 25 mm

H8 **2x**
M3 nut

From LCD **1x**
LCD cable

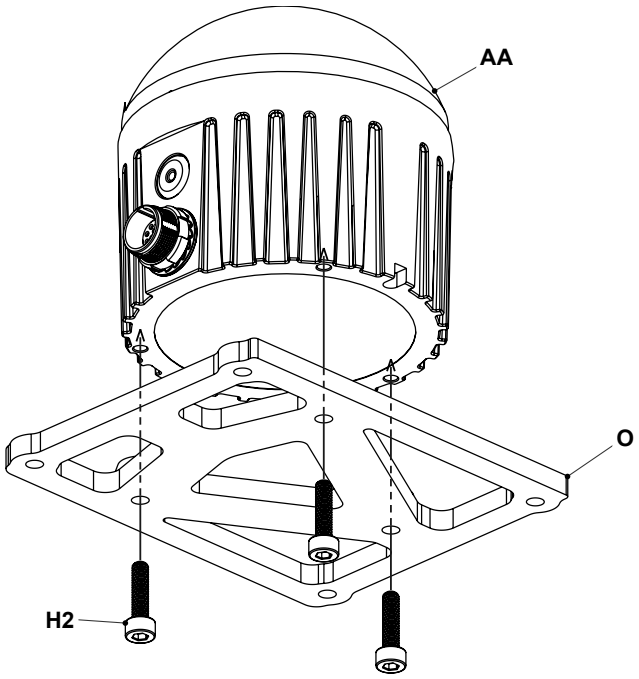
ALREADY ASSEMBLED

Body + camera
From Step 21

LCD assembly
From Step 22

STEP 24

Fasten Airy to the LiDAR mount



Align the three mounting holes in the Airy base with the LiDAR mount. Use three M3 x 12 screws through the plate into the LiDAR. Seat the unit evenly and avoid bottoming the screws.

Keep the LiDAR optical dome and ventilation clear.

PARTS ADDED

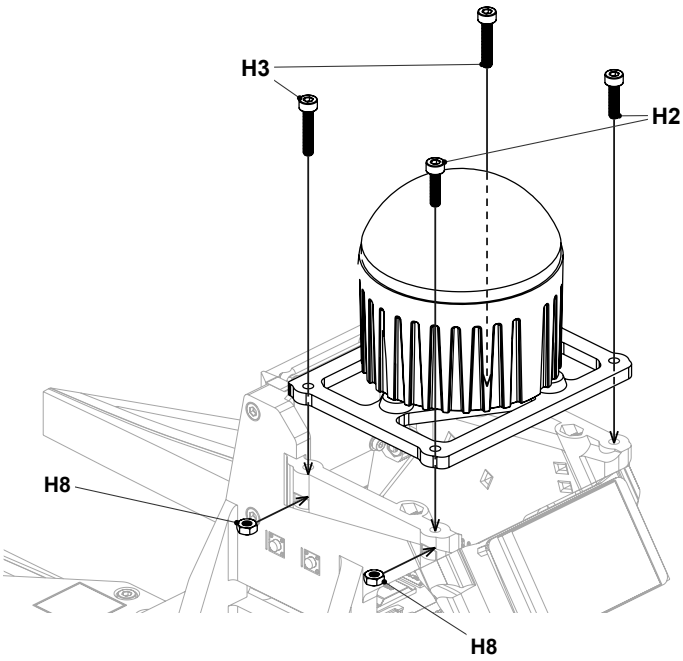
O **1x**
LiDAR mount

AA **1x**
RoboSense Airy LiDAR

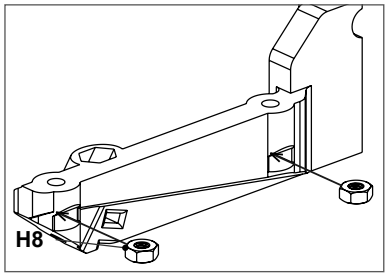
H2 **3x**
M3 × 12 mm

STEP 25

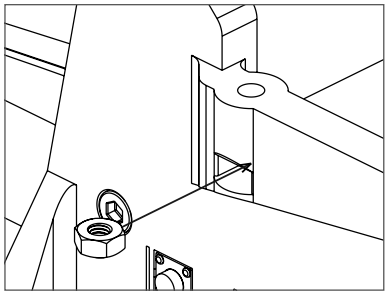
Install the LiDAR assembly



OPPOSITE SIDE / CUTAWAY



VISIBLE SIDE / SLOT DETAIL



Slide two M3 nuts inward from each outer side into the four post slots. Place the LiDAR mount on the supports. Fit M3 x 16 screws at the front holes (camera side) and M3 x 12 screws at the rear (LCD side).

Connect the Airy cable to the LiDAR by hand screwing it in. Then plug one end of the Ethernet cable into the Airy cable and the other end into the Raspberry Pi.

PARTS ADDED

H2 **2x**
M3 × 12 mm

H3 **2x**
M3 × 16 mm

H8 **4x**
M3 nut

AK **1x**
Ethernet cable

From LiDAR **1x**
Interface cable

ALREADY ASSEMBLED

Body
From Step 23

LiDAR assembly
From Step 24

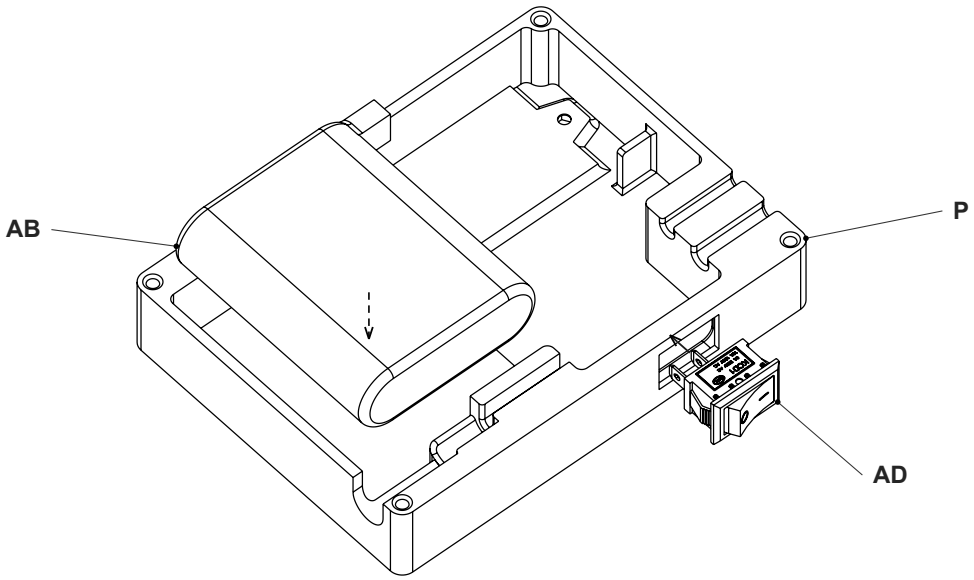
STEP 26

optional cute wire lengths

PLACEHOLDER

STEP 27

Fit the battery and switch



Keep the battery disconnected and the switch OFF. Seat the battery in the large compartment. Push the switch into the side opening until it clips into place.

PARTS ADDED

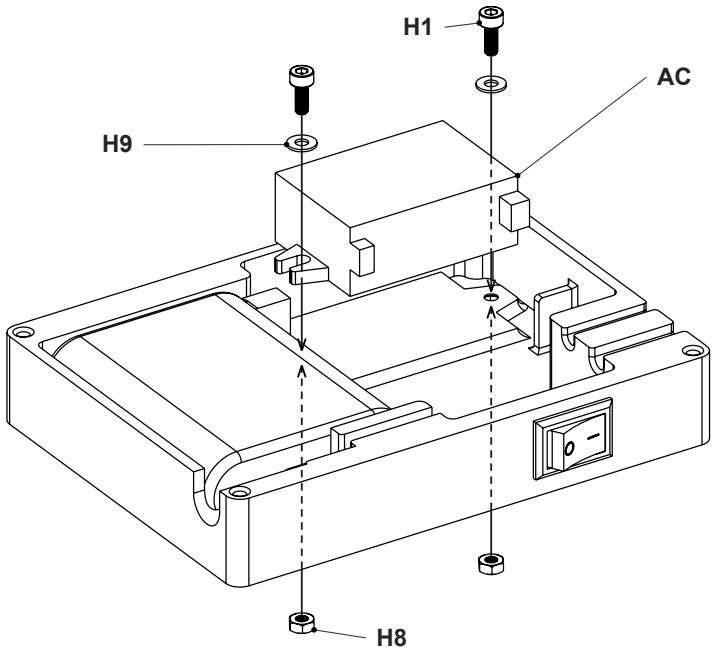
P **1x**
Battery pack bottom

AB **1x**
Battery

AD **1x**
Power switch

STEP 28

Secure the converter



Place the converter beside the battery. Secure it with two M3 x 8 mm screws, placing a washer under each screw head. The two nuts go beneath the case.

PARTS ADDED

AC **1x**
DC-DC converter

H1 **2x**
M3 x 8 mm screw

H8 **2x**
M3 nut

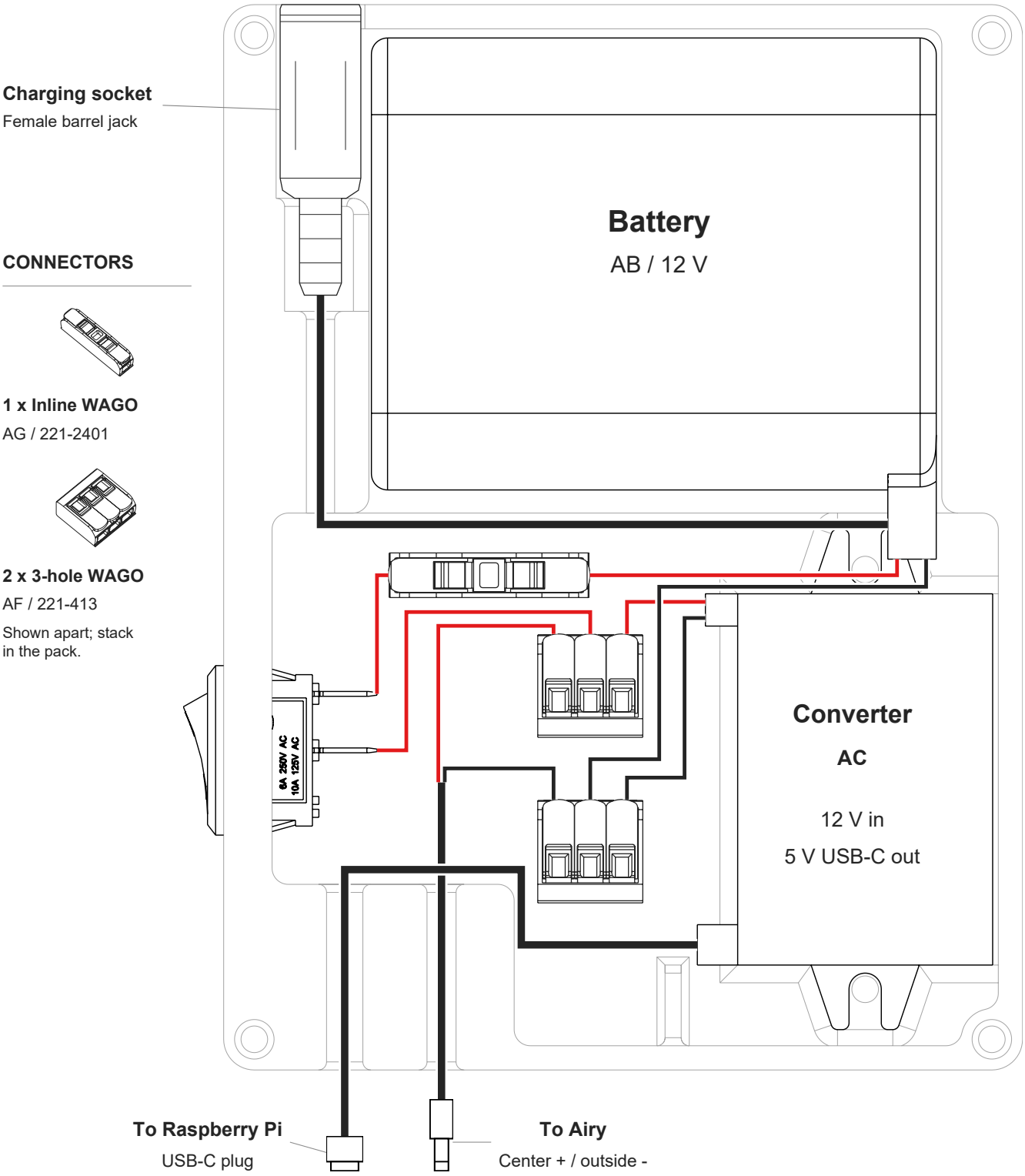
H9 **2x**
M3 washer

ALREADY ASSEMBLED

Open battery pack
From Step 27

STEP 29

Wire the battery pack



Positive (+) Negative (-) Gaps at wire overlaps = no connection.

Connect one wire per hole: strip to the marked length and gather all strands. Optional anti-fraying: solder only the very tip; leave the clamped section unsoldered. Lift the lever, insert fully, close, then gently tug-check. All three holes of each 221-413 connect internally.

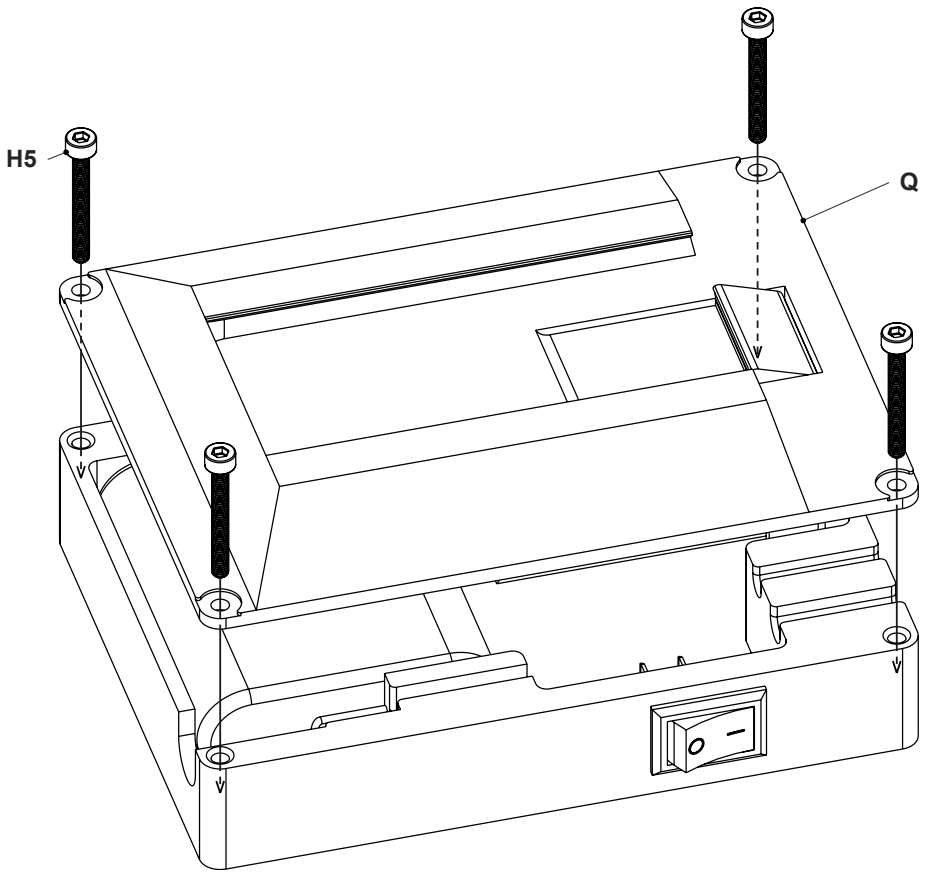
Output test: leave the Pi and Airy unplugged. Connect the battery and switch ON. Set the multimeter to DC volts: red probe in the barrel center, black probe on the outside. Expect about +12 V. Then switch OFF and disconnect the battery.

STEP 30

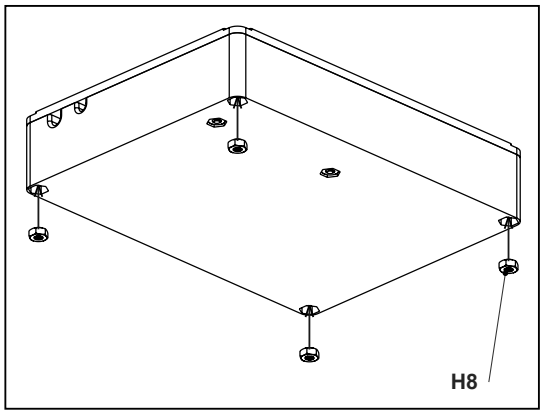
Close the battery pack

Finish the wiring and output test on page 24 before closing the case.

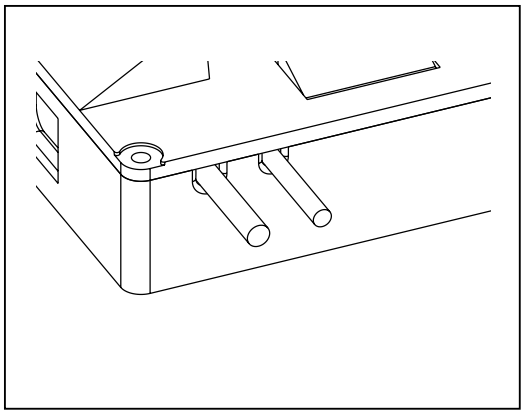
FIT THE COVER



NUT POCKETS / UNDERSIDE



CABLE EXITS / LID SEATED



1. After the output test passes, keep the switch OFF and reconnect the battery. Put each output cable into its case outlet.

2. Lower the lid carefully. Keep every wire away from the lid edge, screw posts and screw holes.

3. Fit four M3 x 25 mm screws and four nuts. Check that the switch is accessible and both cables are held securely.

PARTS ADDED

Q **1x**

Battery pack top

H5 **4x**

M3 x 25 mm screw

H8 **4x**

M3 nut

ALREADY ASSEMBLED

Wired battery pack

From Step 29

STEP 31

Slide on the battery pack

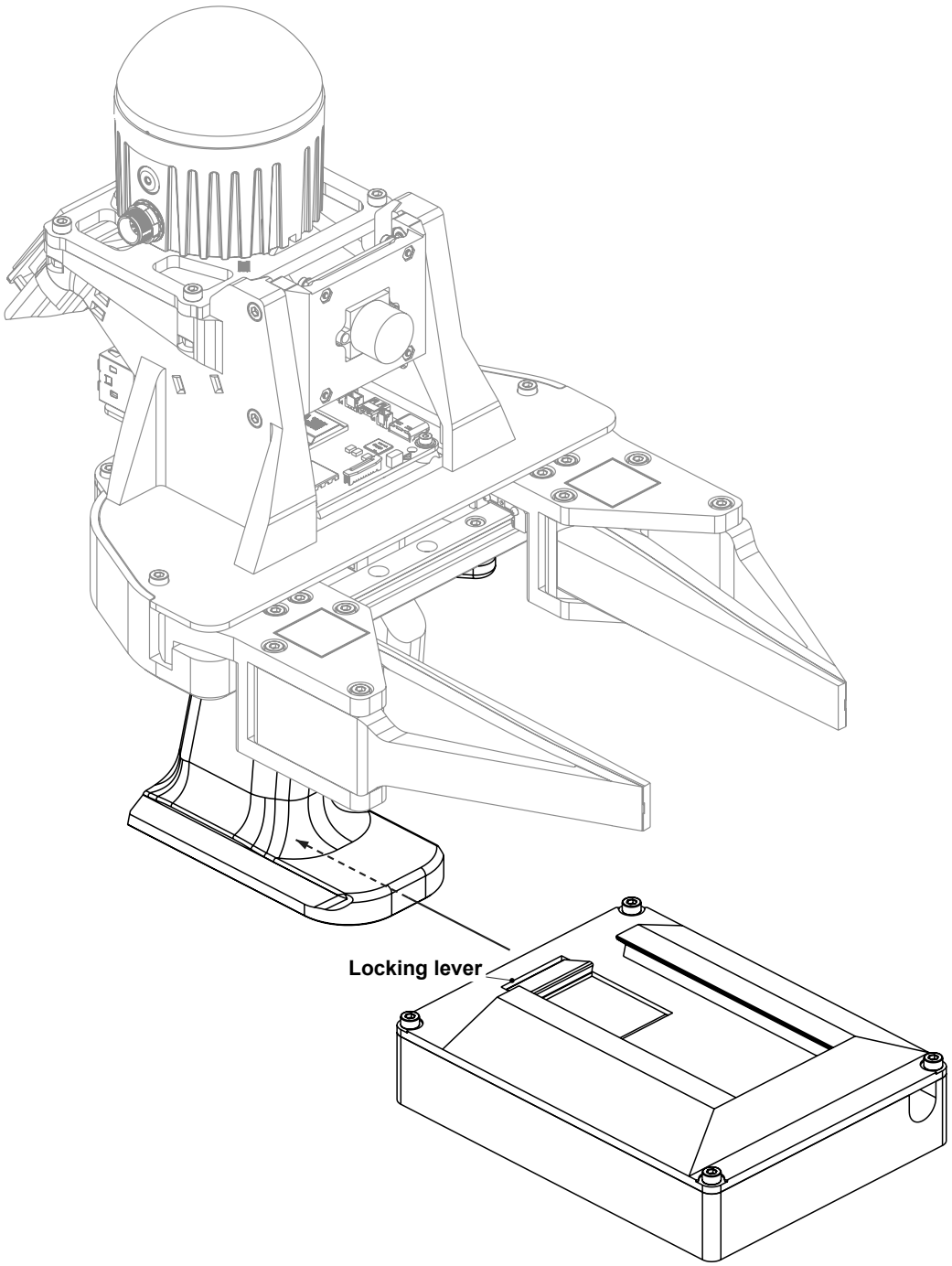
ALREADY ASSEMBLED

UMI-Live assembly

From Step 25

Battery pack

From Step 30

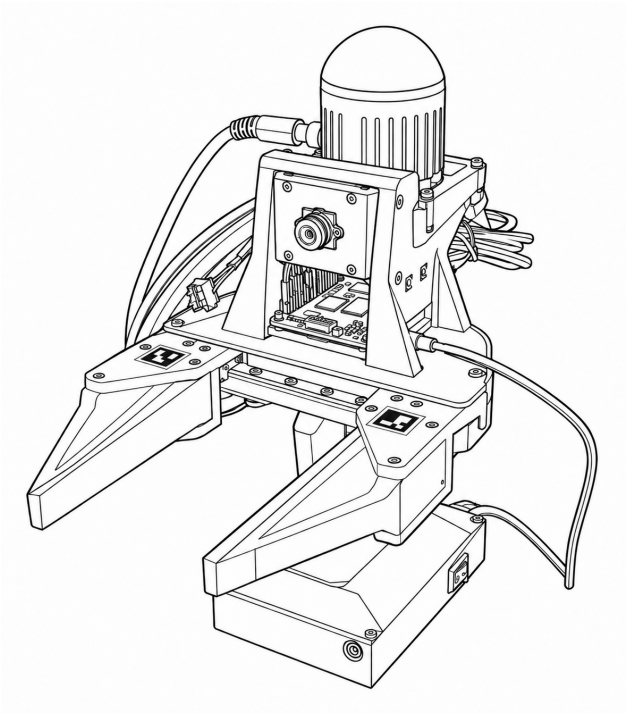


Align the battery-pack rails with the base of the grip.
Slide the pack until it clicks and locks into place.

If the battery pack lever is unable to be pushed down, open the battery pack and ensure that the components inside are not obstructing its motion.

STEP 32

Connect the finished hardware



CABLE ROUTING REFERENCE

Illustrated from the UMI-Live repository photo.

With the switch OFF, connect the pack USB-C output to the Pi power input and its DC barrel output to the Airy power input. Confirm Airy Ethernet goes to the Pi RJ45 port and the camera goes to USB 3.0.

Complete the final checks below before switching ON.

ALREADY ASSEMBLED

Completed hardware

From Step 31

CONNECTIONS

CAMERA

USB 3.0 to Pi

AIRY DATA

Ethernet to Pi

PI POWER

Pack USB-C

AIRY POWER

Pack DC barrel to original cable

Final check

- ☐ Trigger returns under spring force; both fingers move smoothly together.
- ☐ Fasteners are seated; bearings and carriages remain free.
- ☐ Tags are flat, visible and have distinct IDs.
- ☐ Buttons use pins 32/30 and 36/34. LCD follows the pin map.
- ☐ Camera connects to USB 3.0; Airy connects to Ethernet and its original power cable.
- ☐ No bare conductors, trapped wires or cable loops in the moving mechanism.
- ☐ Pack output test on page 24 passed; both assemblies are fully seated.

Charge the battery

1. Switch OFF. Disconnect all connections to the battery, including both bare-wire leads from the WAGO harness.
2. Plug the supplied charger directly into the battery's charging barrel socket.
3. Unplug the charger before reconnecting the battery to the harness.

Do not charge and use the battery at the same time.